

Initial deployments

"802.16m requires new hardware that meets the specifications. However, the standard is fully backward-compatible. So 802.16e equipment can interoperate with 802.16m equipment. This ensures smooth migration paths for 802.16e operators and interoperability in roaming between both types of operators," points out Prateek Kapadia, senior manager, Tata Teleservices Limited (TTSL)-IIT Bombay Center for Excellence in Telecom.

"802.16m chipsets should start becoming available in the 2011-12 timeframe," says Jose Puthenkulam, director of WiMAX Standards in Intel's Mobile Wireless Group. Intel members hold key positions in both the 802.16 Working Group and the 802.16m Task Group.

Multi-carrier modulation amplifiers, radios and channel cards will also require changes. Additional hardware and associated software modifications would be needed in mobile stations and the access service network including the base station, to support Layer 1 and 2 modifications for self reconfigurations, scheduler, security and power control. Minor changes might be required to the connectivity service network.

System profiles for 802.16m-based WiMAX Release 2 are expected to be defined in the fourth quarter of 2010 and certified in 2011. Commercial deployments are expected in the 2011-12 timeframe.

vantages. So which technology will really become the building block of 4G? Or, is this very discussion redundant considering that 802.16e WiMAX itself has not spread its wings as fast as was expected, except in Russia (where service provider Yota has made exemplary use of WiMAX)?

"This is a very interesting, complex and thoughtful question, and the response depends upon a whole range of dynamics involving the existing telecom environment, domineering cellular network providers, handset availability, economy of scale, introduction of disruptive technologies, and timeliness and allocation of spectrum," says Dr Borkar.

In developing countries like India, the primary impetus to initiate the 802.16e WiMAX is to exploit use of the broadband wireless access (BWA) spectrum being allocated soon. This will provide broadband access diffusion and also the required experience for introduction of 802.16m WiMAX, especially for rural and tier-2 cities. Cellular profiles do not exist in these higher frequencies. In the cellular arena, introduction of LTE will be somewhat throttled due to the adequacy of 3G wideband code-division multiple access (WCDMA) and high-speed packet access (HSPA) networks in meeting the application requirements

in the foreseeable future.

In developed countries with mature stationary, nomadic and cellular broadband environments and highly developed infrastructure in place, dominant cellular providers have already introduced extensive 3G WCDMA/HSPA and CDMA2000 networks and are putting plans in place to migrate to LTE some time in future. User handsets are relatively affordable and widely available. In such environments, 802.16e WiMAX has limited applicability and smaller players like Clearwire and Sprint have had limited success.

Also, a key disadvantage for WiMAX has been the necessity to use comparatively higher carrier frequencies. At higher carrier frequency, a larger number of base stations are required for the same geographical coverage. For example, going from 2.5 GHz to 3.5 GHz doubles the number of base stations required. In the US, even when the attractive 700/800MHz analogue TV frequency spectrum was released and auctioned last year, the already dominant cellular players AT&T and Verizon acquired a major share resulting in continuing cellular dominance.

"Developing countries like India are still in the 2G/2.5G cellular stage with limited infrastructure in place. With basically a non-existent 3G

network, WiMAX has a major opportunity even if the BWA spectrum is significantly higher than the 3G carrier. 802.16e WiMAX may be considered a complement to 3G and is a good candidate for overall broadband diffusion in providing fixed and nomadic broadband access. Moving forward, second-tier cities and rural areas will be the primary target for rapid growth of WiMAX. In these environments, there is substantially less competition from incumbent 3G suppliers. If and when lower frequencies (500-800 MHz) get assigned in India, WiMAX may have another opportunity if the government decides to make the spectrum technology neutral," continues Dr. Borkar.

Hence 802.16e WiMAX has its competitive space in networks in developing countries and is likely to be the only viable broadband wireless option in the high-BWA spectrum. It will form the experience base for moving to 802.16m networks. WCDMA/HSPA will continue to be the dominant 3G technologies for the foreseeable future worldwide. 802.16e followed by 802.16m WiMAX is likely to be introduced in developing countries ahead of LTE. LTE will be a clear 4G candidate in the 2012+ timeframe, albeit with disruptive upgrades to 3G WCDMA/HSPA networks. Because of the inevitability of 802.16m WiMAX networks prior to that, dual-mode handsets will become the market requirement to handle both WiMAX and LTE networks and will neutralise the handset advantage for 3G and LTE.

Alright then! Whether as 802.16m or as LTE Release 10, or a combination of both, 4G is going to be here in a few years. So will India go fast on 3G or again skip a generation and jump straight to 4G? "No, we can not wait. We must move on to 3G today and upgrade to 4G in two years. Anyway, BWA bidding is such that it can be upgraded to 4G," signs off Dr Jhunjhunwala. ●

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